Fiscal Year:	FY 2014	**Task Last Updated:**	FY 08/03/2013
PI Name:	Palinkas, Lawrence Ph.D.		
Project Title:	Assessing the Impact of Communication Delay on Performance: An Examination of Autonomous Operations Utilizing Analog Environments		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Los Angeles	**State:**	CA
Zip Code:	90089-0411	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/03/2011	**End Date:**	06/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Leveton, Lauren	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date is questionable as study was complete in 2012 (Ed., 8/6/13)		
Key Personnel Changes/Previous PI:	NOTE: Original PI was K. Keeton, but replaced by L. Palinkas before project initiated, per M. Covington/JSC HRP (Ed., 5/31/2012)		
COI Name (Institution):	Schmidt, Lacey (NASA Johnson Space Center/Wyle) Slack, Kelley (NASA Johnson Space Center/Wyle) Malka, Ari (NASA Johnson Space Center/Wyle) Baskin, Pamela (NASA Johnson Space Center/Wyle) Cockrell, Gabrielle (NASA Johnson Space Center/Wyle) Ploutz-Snyder, Rob (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The NASA/JSC Behavioral Health and Performance Element (BHP) of the Space Medicine Division, Human Research Program, is proposing a research study that will examine what impact implementing a communication delay will have on individual and team outcomes, including performance. To date, very few studies have observed teams in environments without communication to ground control, and no such studies have been conducted during long-duration missions. This preliminary study took place in an analog environment (NEEMO) and provided an opportunity to compare team performance across varying amounts of communication delays, using tasks that differ in criticality and novelty. Additionally, the findings of this study were used to inform the second phase of this project, which will take place on ISS. Specifically, this study examined how teams interact and perform tasks with and without communication delays. The tasks performed by the crew varied along two dimensions: 1) tasks that are either critical or not critical (criticality) and 2) tasks that are either novel or familiar (novelty). Tasks were varied by criticality and novelty as it is assumed that highly novel and highly critical tasks are similar to tasks that a crew may encounter during a long duration mission in which they have no prior training but must address. This feasibility study provided a preliminary understanding of what impact communication delay has on individual and team performance as well as insight into how teams perform and interact in high-fidelity, autonomous environments.
Rationale for HRP Directed Research:	This research is directed because NASA must define complete scientific activities in a short time and there is insufficient time to issue a solicitation.
Research Impact/Earth Benefits:	The research provided insight into performance associated with asynchronous communications across distributed groups comprising a team in any location.
Task Progress:	Data collection was completed on June 15, 2012 and study results were reported at the annual meetings of the NASA Johnson Space Center Behavioral Healh and Performance Working Group. Transition to Study using the International Space Station: In the past year, lessons learned from the NEEMO studies were incorporated into the design of a protocol for data collection to take place aboard the International Space Station. The protocol was submitted to the Astronaut Office and to the Human Research Program for review and approval. Much of the past year was devoted to recruitment in Spring, 2014.
Bibliography Type:	Description: (Last Updated: 11/13/2019)